

Lecture 4: Chemical Differentiation & Magnetospheres

Planetary Systems

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Learning objectives

By the end of this lecture, you will be able to:

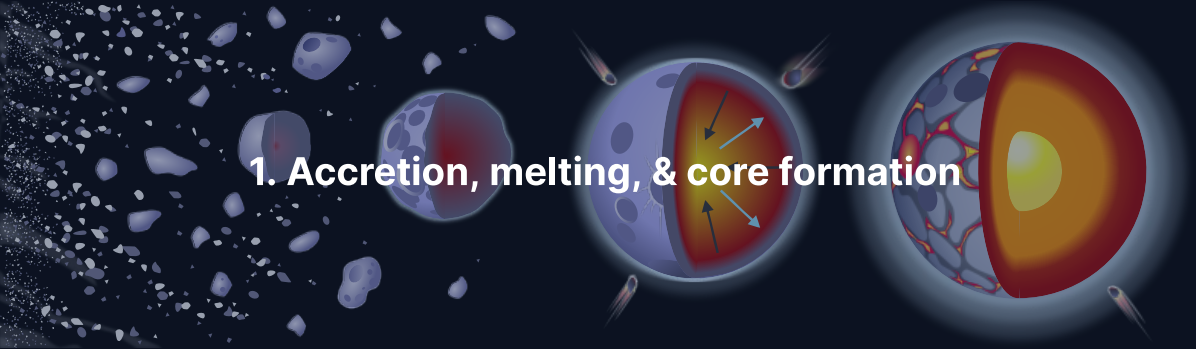
- Explain metal–silicate separation and core formation
- Use siderophile element depletion as evidence for differentiation
- Describe the Hf–W chronometer for dating core formation
- Derive the magnetic Reynolds number and assess dynamo feasibility
- Compare magnetic fields across the solar system

Recap: Lecture 3

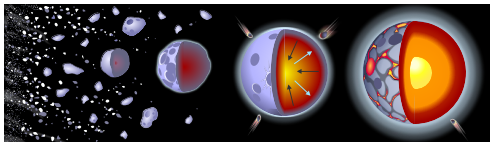
- Four heat sources: accretional, differentiation, radioactive, tidal
- Conduction too slow for planets: $\tau \sim L^2/\kappa$
- Rayleigh number: $Ra \sim 10^7\text{--}10^8$ for Earth $\Rightarrow$ vigorous convection
- Plate tectonics (mobile lid) vs. stagnant lid
- Tidal heating: Io, Enceladus, Europa

Today: The *chemical* consequences of melting, and how differentiation enables magnetic fields.

1. Accretion, melting, & core formation



Metal-silicate separation

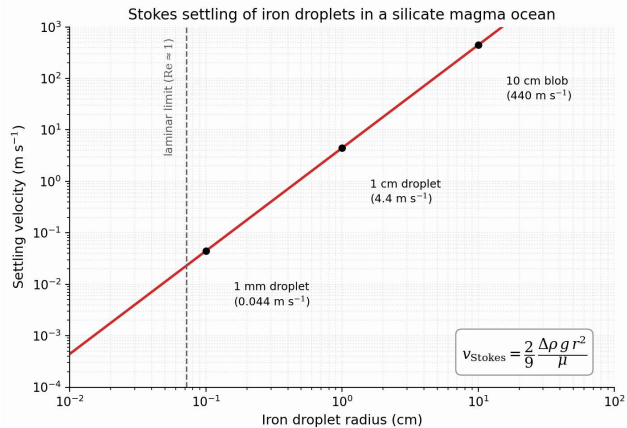


Credit: Wikimedia, CC BY-SA 4.0

- Accretional heating + $^{26}\text{Al} \rightarrow$ **magma ocean**
- Iron ($\rho \sim 7000 \text{ kg m}^{-3}$) is immiscible with silicate ($\rho \sim 3000 \text{ kg m}^{-3}$)
- Iron droplets sink through silicate melt
- **Stokes settling velocity:**

$$v_{\text{Stokes}} = \frac{2}{9} \frac{\Delta \rho g r^2}{\mu}$$

- cm droplets: $v \sim 4 \text{ m s}^{-1}$ (Stokes),
 $\sim 0.5 \text{ m s}^{-1}$ with turbulent drag



Stokes settling of iron droplets in a magma ocean: $v \propto r^2$ (slope 2 on log-log axes), 0.044 m s^{-1} at 1 mm and 4.4 m s^{-1} at 1 cm. Left of the dashed line ($\text{Re} \approx 1$) Stokes' law holds; to its right the flow is turbulent and a 1 cm droplet moves nearer 0.5 m s^{-1} , still crossing a 1000 km magma ocean within weeks. Iron rain empties a magma ocean of its metal within weeks: on geological clocks, core formation is instantaneous. Course figure.

The magma ocean state, and why it is essential

After accretion the outer few thousand kilometres of a terrestrial planet are molten: depth several hundred to >2000 km, surface temperature $2000\text{--}4000$ K, viscosity $10^{-1}\text{--}10^0$ Pa s, turbulent convection ($Ra \gtrsim 10^{25}$), cooling set by surface radiation and greenhouse blanketing.

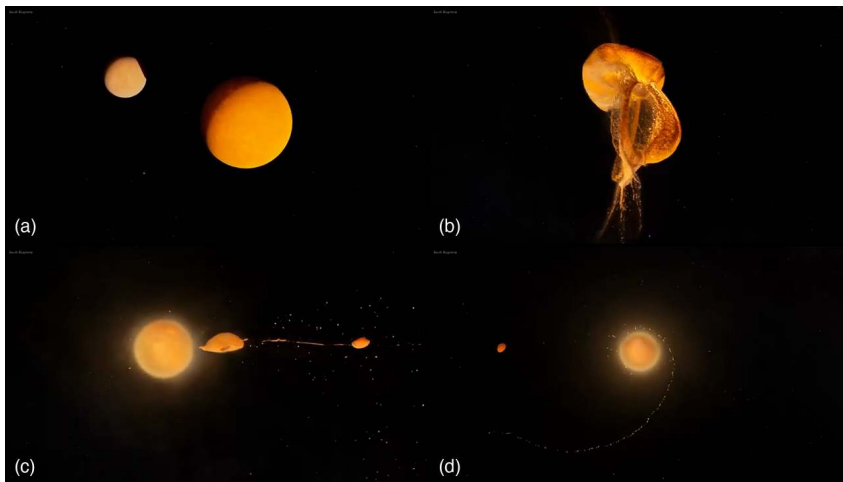
- **Liquid mantle:** iron droplets sink at m s^{-1} , a core forms within weeks
- **Solid mantle:** rock flows so slowly ($\mu \sim 10^{21}$ Pa s, 22 orders of magnitude above silicate melt) that even a km-sized iron body would need longer than the age of the universe to reach the centre
- Magma ocean lifetime $10^3\text{--}10^8$ yr: long enough for core formation, short compared to planetary evolution

Core formation happens while the planet is molten, or not at all; chondritic asteroids that never melted never formed cores.

The Moon-forming giant impact

- Mars-sized body (Theia) collided with proto-Earth ~4.5 Ga
- Re-melted most of Earth's mantle, vaporised part of it
- Created the final deep magma ocean
- Established present core-mantle structure
- Debris disk → Moon (confirmed by lunar $\epsilon^{182}\text{W}$ matching Earth's mantle)

Core formation is not a single event but a series of episodes, culminating in the giant impact.



SPH simulation of the Moon-forming giant impact: (a) Mars-sized Theia approaches the proto-Earth, (b) contact and tidal disruption, (c) part of the debris falls back while outer clumps survive, (d) a satellite on a wide orbit. Kegerreis et al. (2022).

► Play animation

► Source video

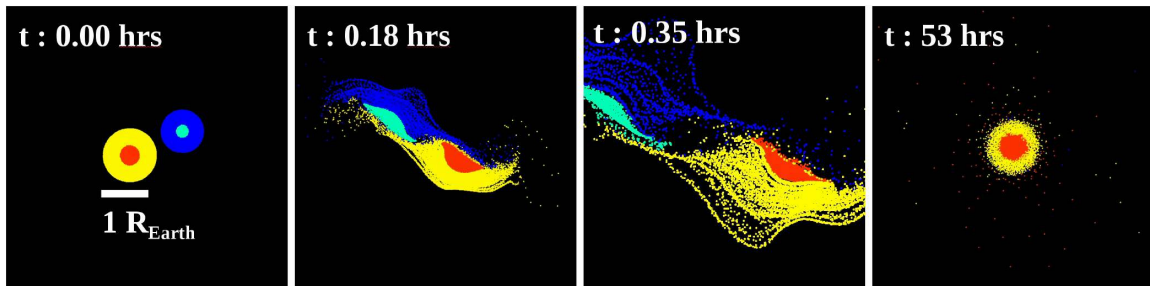
Giant impacts that remove mantles

Not every giant impact ends in net accretion: an oblique, fast collision can strip more silicate than it delivers.

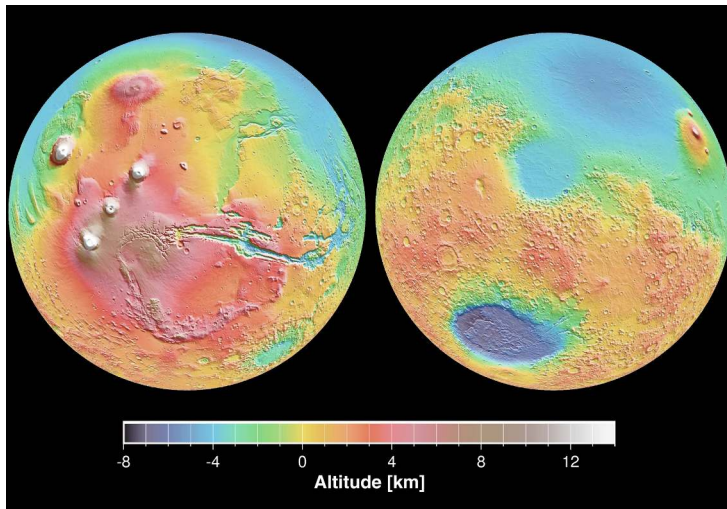
- **Mercury:** core mass fraction ~74% (Earth ~32%, Mars ~24%); a hit-and-run impact (an oblique collision whose remnant re-accretes rather than merging) stripped ~50–70% of the pre-impact mantle
- **Mars:** the hemispheric dichotomy (~6 km elevation, ~30 km crustal-thickness contrast) is the relaxed scar of a Borealis-scale impact; crust moved sideways, not lost
- **Earth–Moon:** the opposite endpoint, net accretion of the impactor

Mercury and Earth are the two endpoints of the same process: giant impacts set the final core-to-mantle ratio.

Source: Benz et al. (2007); Chau et al. (2018); Andrews-Hanna et al. (2008); Marinova et al. (2008)



A mantle-stripping giant impact on proto-Mercury (SPH simulation): a proto-Mercury of 2.25 present Mercury masses (red core, yellow mantle) is hit by an impactor of half its mass at impact parameter $b = 0.5$ and $v = 30 \text{ km s}^{-1}$; the white bar marks $1 R_{\oplus}$. Silicate is ejected while the metal-rich remnant recollapses; after 53 hours the largest fragment holds 0.95 Mercury masses with iron mass fraction $Z_{\text{Fe}} = 0.67$. One oblique collision at several escape velocities converts an ordinary protoplanet into iron-rich Mercury. Chau et al. (2018).



Mars's hemispheric dichotomy in MOLA global topography: smooth, low northern plains (blue) against cratered southern highlands (orange to red); the deep blue basin on the right is Hellas. Credit: NASA/JPL.

The late veneer

Highly siderophile elements (Ir, Pt, Os, Au) are **overabundant** in Earth's mantle compared to the deep-core equilibration prediction.

- Mantle abundances $\sim 200\times$ higher than expected for full metal extraction
- Relative ratios are **chondritic** (matching primitive meteorites), not fractionated
- Interpretation: an extra $\sim 0.5\%$ of Earth's mass accreted **after** core formation closed
- This “late veneer” delivered its siderophile elements to the mantle without equilibrating with the core

Source: Walker (2009); Day et al. (2016)

A smaller late veneer is also inferred for Mars and the Moon from their platinum-group element abundances.

Goldschmidt classification

1 1.008 H hydrogen	2 6.94 Li lithium	3 9.012 Be beryllium	4 12.01 B boron	5 10.81 C carbon	6 14.01 N nitrogen	7 16.00 O oxygen	8 19.00 F fluorine	9 20.18 Ne neon	10 22.99 Na sodium	11 24.31 Mg magnesium	12 26.98 Al aluminum	13 28.09 Si silicon	14 29.89 P phosphorus	15 32.07 S sulfur	16 35.45 Cl chlorine	17 39.95 Ar argon	18 40.08 K potassium	19 44.96 Ca calcium	20 47.88 Sc scandium	21 50.94 Ti titanium	22 52.00 V vanadium	23 54.94 Cr chromium	24 55.85 Mn manganese	25 58.93 Fe iron	26 58.93 Co cobalt	27 58.93 Ni nickel	28 63.55 Cu copper	29 65.39 Zn zinc	30 69.72 Ga gallium	31 72.64 Ge germanium	32 74.92 As arsenic	33 78.96 Se selenium	34 79.90 Br bromine	35 83.80 Kr krypton	36 85.47 Rb rubidium	37 87.62 Sr strontium	38 88.91 Y yttrium	39 91.22 Zr zirconium	40 92.91 Nb niobium	41 95.94 Mo molybdenum	42 98.91 Tc technetium	43 101.07 Ru ruthenium	44 101.07 Rh rhodium	45 106.42 Pd palladium	46 107.87 Ag silver	47 112.41 Cd cadmium	48 112.41 In indium	49 114.82 Sn tin	50 118.71 Sb antimony	51 127.60 Te tellurium	52 127.60 I iodine	53 131.29 Xe xenon	54 132.91 Cs cesium	55 137.33 Ba barium	56 137.33 La lanthanum	57 140.91 Ce cerium	58 140.91 Pr praseodymium	59 140.91 Nd neodymium	60 140.91 Pm promethium	61 140.91 Sm samarium	62 150.36 Eu europium	63 151.96 Gd gadolinium	64 157.25 Tb terbium	65 158.93 Dy dysprosium	66 162.50 Ho holmium	67 164.93 Er erbium	68 167.26 Tm thulium	69 168.93 Yb ytterbium	70 173.05 Lu lutetium	71 175.04 Ac actinium	72 175.04 Th thorium	73 175.04 Pa protactinium	74 175.04 U uranium	75 175.04 Np neptunium	76 175.04 Pu plutonium	77 175.04 Am americium	78 175.04 Cm curium	79 175.04 Bk berkelium	80 175.04 Cf californium	81 175.04 Es einsteinium	82 175.04 Fm fermium	83 175.04 Md mendelevium	84 175.04 No nobelium	85 175.04 Lr lawrencium	86 175.04 Uuo ununoctium	87 175.04 Uus ununseptium	88 175.04 Uuh ununhexium	89 175.04 Uuo ununoctium	90 175.04 Uus ununseptium	91 175.04 Uuh ununhexium	92 175.04 Uuo ununoctium	93 175.04 Uus ununseptium	94 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Partition coefficients, and why silicon is in the core

$$D_i^{\text{met/sil}} \equiv C_i^{\text{metal}} / C_i^{\text{silicate}}$$

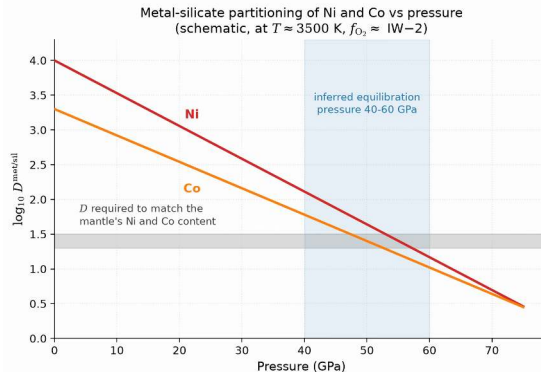
Element	$D^{\text{met/sil}}$	Behaviour
Ir, Os, Pt	10^4 – 10^6	Highly siderophile
W, Mo	10^2 – 10^4	Moderately siderophile
Ni, Co	10–100	Weakly siderophile
Fe	~10	Weakly siderophile
Si, Mg, O	$\ll 1$	Lithophile

D depends on pressure, temperature and oxygen fugacity (an effective O_2 partial pressure that sets the redox state of the melt):

- Earth's mantle is depleted $\sim 1000\times$ in Ir and Pt: the iron-loving elements followed the metal into the core
- Si is lithophile below 10 GPa, but D_{Si} rises to ~ 1 at 40–60 GPa: the core holds several wt% Si, part of its 5–10% density deficit (Lecture 8)

Deep magma ocean partitioning at 40–60 GPa is the standard model for Earth's core composition.

Source: Rubie et al. (2015); Fischer et al. (2015); Hirose et al. (2021); Shahar et al. (2026)



Metal-silicate partition coefficients against pressure: laboratory $D^{\text{met/sil}}$ for Ni and Co at magma-ocean conditions. At low pressure $D \sim 10^3$ – 10^4 , so the mantle should keep almost no Ni or Co; the grey band ($D \approx 24$ – 26) is the value required by the Ni and Co the mantle actually holds. Both curves reach the band only near 40–60 GPa: equilibration at 54 ± 5 GPa, the base of a 1000–1500 km deep magma ocean (Fischer et al. 2015). The mantle's siderophile depletions remember the depth at which the core formed. Course figure, after Siebert et al. (2013).

Hf–W chronometry: when did cores form?

$^{182}\text{Hf} \rightarrow ^{182}\text{W}$, half-life 8.9 Myr. Hf is lithophile (stays in the mantle), W is siderophile (goes to the core). Early core formation strips W while ^{182}Hf is still alive, so the mantle builds an excess of ^{182}W ; late core formation (after ^{182}Hf is gone) leaves the mantle chondritic, $\epsilon^{182}\text{W} \approx 0$.

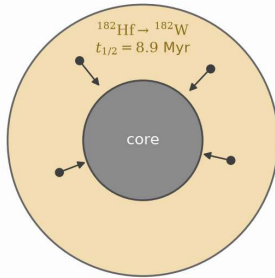
Body	$\epsilon^{182}\text{W}$	Core formation timing
Earth	+2	two-stage age ~30 Myr, a lower limit
Mars	+0.4 bulk, +1.8 shergottites	complete within ~10 Myr
Moon	~Earth	consistent with the giant impact

Earth's bulk core formation finished within ~100 Myr; depleted shergottites are a martian meteorite class.

Source: Kleine et al. (2009); Nimmo & Kleine (2015); Kleine & Walker (2017)

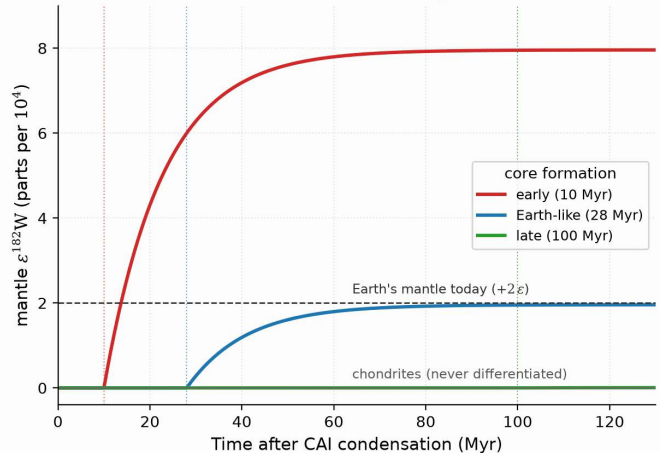
(a) parent and daughter separate

mantle: Hf (lithophile) stays

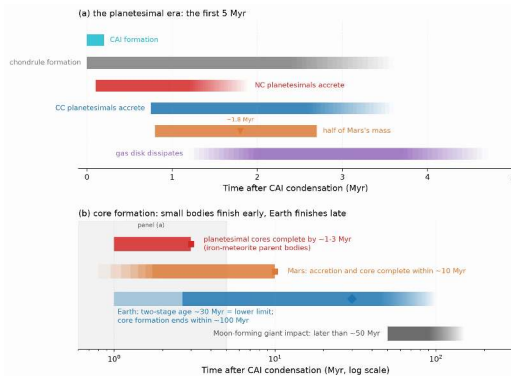


W (siderophile)
sinks into the core

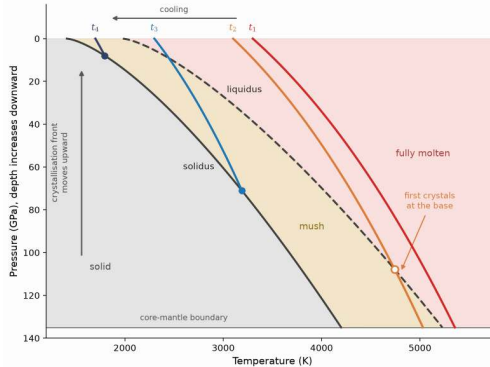
(b) earlier core formation, larger excess



(a) Core formation separates parent (Hf, mantle) from daughter reservoir (W, core). (b) Mantle ^{182}W excess vs. time of core formation, two-stage model with Earth's mantle Hf/W: the measured $+2\epsilon$ gives ~ 28 Myr. Course figure.



(a) The first 5 Myr after CAI condensation (CAIs, calcium-aluminium-rich inclusions, the first nebular solids, define $t = 0$): chondrules and planetesimals within ~4 Myr (Lecture 12). (b) Core formation to 300 Myr: planetesimal cores (iron meteorites) by ~1–3 Myr, Mars within ~10 Myr, Earth's two-stage age of ~30 Myr as a lower limit with completion within ~100 Myr, the Moon-forming impact after ~50 Myr. Small bodies differentiate within 1–3 Myr; Earth needs tens of Myr. Course figure, after Lichtenberg et al. (2023); ages from Nimmo & Kleine (2015), Kleine & Walker (2017), Kruijer et al. (2017).



Peridotite solidus and liquidus against magma-ocean adiabats at successive times t_1 – t_4 . The melting curves rise more steeply with pressure than the adiabats, so the cooling adiabat first meets the liquidus at the base (t_2 , open circle), where bridgmanite (MgSiO_3) crystallises first; the adiabat-solidus intersection (dots) is the crystallisation front, shallower at each later time; between the curves crystals and melt coexist as mush. Caveat: dense Fe-rich melt can survive at the base as a basal magma ocean (Labrosse et al. 2007; Samuel et al. 2023). Deep magma oceans crystallise from the bottom up. Course figure, after Elkins-Tanton (2012).

Incompatible elements and mantle reservoirs

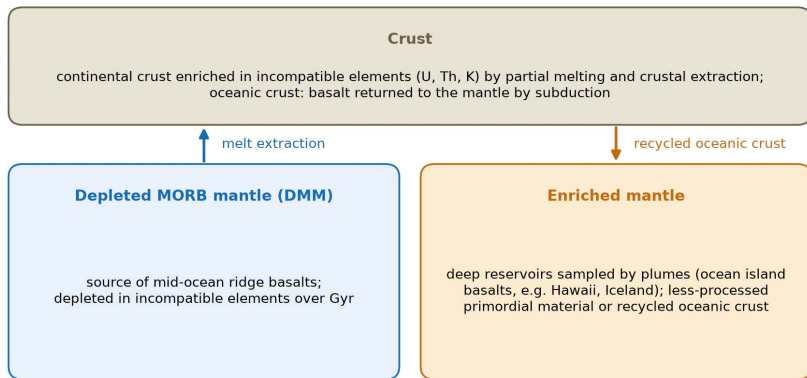
Large ions (U^{4+} , Th^{4+} , K^+) fit poorly into crystal lattices, concentrate in the shrinking residual liquid and end up in the crust: continental crust is 50–100× enriched in U, Th and K over the upper mantle, and about half of Earth's radiogenic heat comes from a crust that is <1% of its mass.

DMM Depleted MORB Mantle: upper-mantle source of mid-ocean ridge basalts, depleted by partial melting and crustal extraction.

EM/OIB Enriched Mantle: deep plumes sample primordial mantle or recycled oceanic crust with preserved incompatible elements.

HIMU “High- μ ” reservoir: high U/Pb, likely very old subducted oceanic crust.

Isotopic diversity of ocean island basalts (Hawaii, Iceland, Samoa) is direct evidence for long-lived mantle heterogeneity.



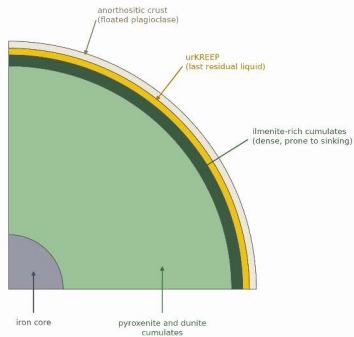
Mantle reservoirs after magma ocean crystallisation and later melting. The depleted MORB mantle, source of mid-ocean ridge basalts, lost its incompatible elements to the continental crust. The enriched mantle sampled by plumes, source of ocean island basalts, holds primordial material or recycled oceanic crust. Course schematic.

The lunar magma ocean: direct evidence

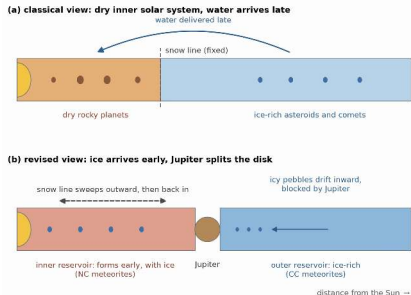
The Moon shows the clearest record of magma ocean differentiation:

- The concept was born from Apollo 11: anorthosite fragments in the Mare Tranquillitatis soil led Wood et al. (1970) to propose a deep molten shell
- **Anorthositic highlands:** low-density plagioclase feldspar floated to the top of the crystallising lunar magma ocean ~4.4 Ga
- Apollo samples of anorthosite date to 4.36–4.44 Ga (within 200 Myr of solar system formation)
- **KREEP** layer: the last dregs of crystallisation, rich in **K**, **R**are **E**arth **E**lements, **P**hosphorus; shallow under the nearside maria
- Lunar crust thickness varies from ~30 km (nearside) to ~70 km (farside)

The Moon is the textbook case of magma ocean differentiation; its exposed anorthosite confirms the theoretical crystallisation sequence.



End-state of the lunar magma ocean (thin layers exaggerated): the 40–50 km anorthosite flotation crust, beneath it the yellow urKREEP layer, the last ~1% of liquid to solidify; light green pyroxenite and dunite cumulates; the dark green ilmenite-rich cumulates formed late in the sequence and are denser than everything below, so they sink (mantle overturn) in the first ~500 Myr; an iron core at the centre. Apollo samples match this stratigraphy in detail: the Moon is the clearest test case of magma-ocean theory. Course figure, after Elkins-Tanton (2012).



Classical versus revised picture of volatile delivery. (a) **Classical**: a fixed snow line keeps the inner disk dry, and water arrives late by asteroids and comets scattered inward. (b) **Revised**: the snow line (the radius beyond which water condenses as ice) sweeps outward, then retreats; the earliest inner planetesimals form beyond it and accrete ice (blue bodies), and Jupiter's growth blocks the inward pebble drift and splits the disk into the NC and CC reservoirs. The protoplanets then dry out again by ^{26}Al melting, impact devolatilisation and pebble ablation in primordial atmospheres; most of Earth's remaining water was present at accretion, not delivered afterwards. Course figure, after Alexander et al. (2019), Lichtenberg et al. (2021), Grewal et al. (2024).

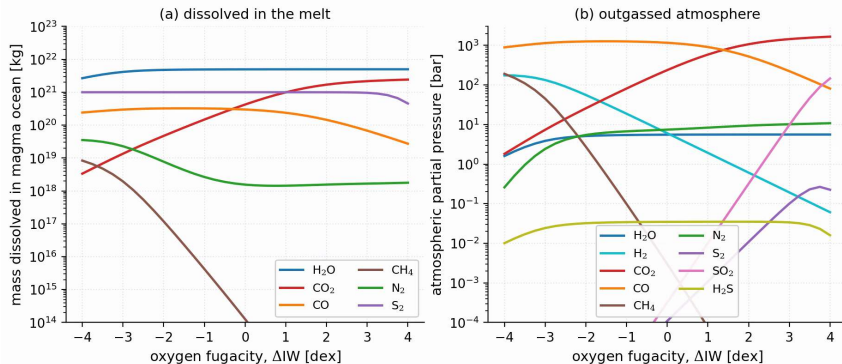
Outgassing and volatile solubility

As the magma ocean cools, dissolved volatiles outgas; speciation follows the **oxygen fugacity** f_{O_2} of the mantle: reducing (low f_{O_2}) gives H_2 , CO , N_2 , CH_4 ; oxidising (high f_{O_2}) gives H_2O , CO_2 , N_2 , SO_2 .

Each gas dissolves through its own reaction with the melt: H_2O as hydroxyl groups, CO_2 as carbonate ions (~100× less soluble), N as nitrides only when reduced, S as sulfide or sulfate. Each water molecule splits into two OH groups:

$$X_{\text{H}_2\text{O}} \propto p_{\text{H}_2\text{O}}^{1/2}$$

A thicker steam atmosphere lets the magma retain more water: a self-limiting feedback, so a planet cannot outgas its entire water budget. The mantle redox state at solidification sets the initial atmosphere (Lectures 5–6). Source: Hirschmann (2012)



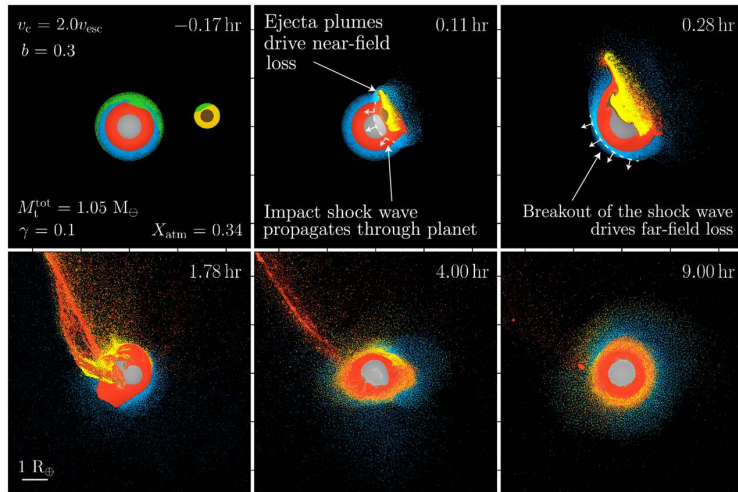
Volatile partitioning between magma ocean and atmosphere against oxygen fugacity (equilibrium outgassing model of Nicholls et al. 2024, bulk silicate Earth inventory of Krijt et al. 2023). (a) The melt keeps almost the whole water inventory dissolved ($\sim 5 \times 10^{21}$ kg, over 3 present-day oceans) at all but the most reducing conditions. (b) Carbon is far less soluble: CO_2 and CO fill the atmosphere at hundreds to thousands of bar while steam stays at a few bar; H_2 , CO and CH_4 when reduced, CO_2 then SO_2 when oxidised. During the magma ocean stage most of the water sits in the magma, not in the atmosphere.

Impact erosion and delivery

- **Local:** an ordinary impactor removes at most the air above the tangent plane, a fraction $H/2R \approx 6 \times 10^{-4}$ of Earth's atmosphere per event; erosion is cumulative, and km-scale planetesimals erode most efficiently per unit mass
- **Global:** a giant impact shakes the whole planet and air is lost wherever the ground approaches v_{esc} ; even the Moon-forming impact removes only ~20%, more with a surface ocean
- The same planetesimals also deliver volatiles; Venus keeps ~50× more ^{36}Ar than Earth because oceans on proto-Earth enhanced its loss
- A primordial H_2 -He envelope dies by attrition: repeated impacts plus XUV-driven thermal escape (Lecture 5)

Earth's air and oceans are the ledger of delivery minus erosion, closed by magma-ocean outgassing.

Source: Melosh & Vickery (1989); Genda & Abe (2003, 2005); Schlichting et al. (2015)



Atmospheric loss in a giant impact: ejecta plumes drive near-field loss at the impact site, while the shock breaking out on the far side drives far-field loss; green particles are lost.

Roche et al. (2025), CC BY 4.0.

Meteoritic dichotomy: NC vs. CC

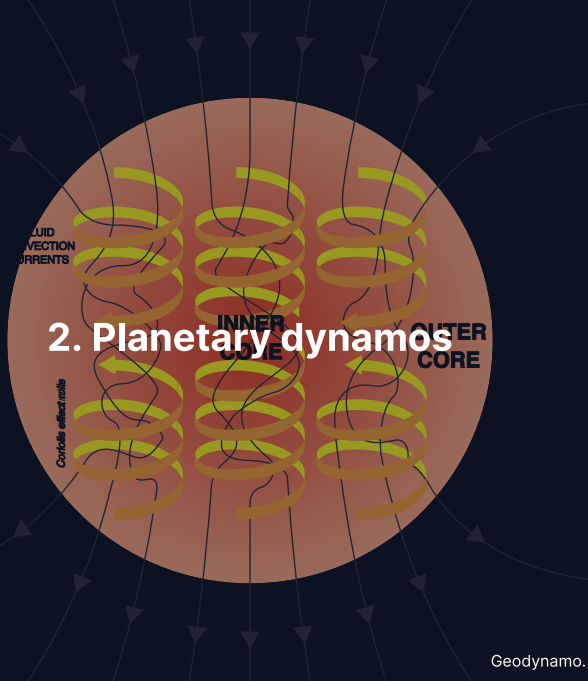
Meteorites split into two isotopic groups: **non-carbonaceous (NC)** and **carbonaceous (CC)**.

- NC and CC have distinct ^{50}Ti , ^{54}Cr , ^{48}Ca ratios
- They reflect two separate reservoirs in the early solar nebula
- Interpretation: **Jupiter** formed early, blocking mixing between inner and outer disk
- Both reservoirs contributed to Earth's accretion, but in changing proportions over time

Source: Kruijer et al. (2017); Lichtenberg et al. (2021)

This dichotomy is one of the strongest meteoritic constraints on the timing of giant planet formation.

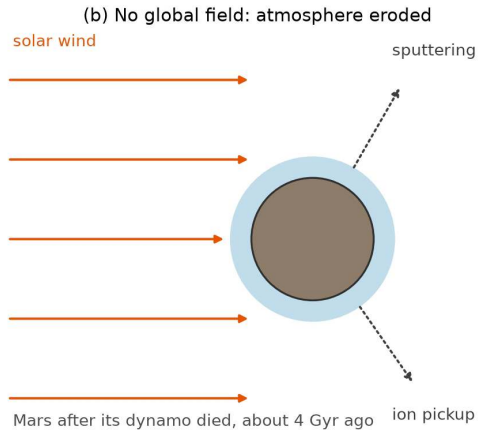
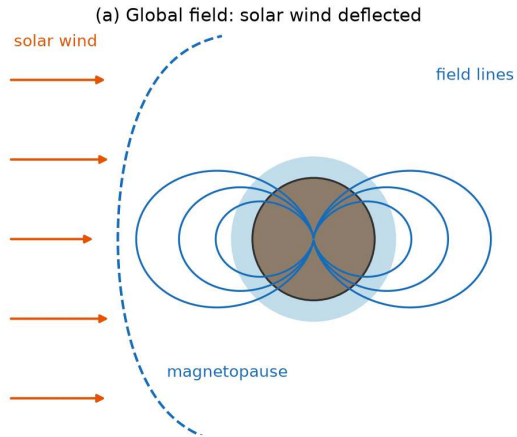
2. Planetary dynamos



Why magnetic fields matter

- **Atmospheric shielding:** deflects the solar wind and reduces ion sputtering (atoms knocked loose by particle impacts) of the upper atmosphere; Mars lost its field ~4 Ga and then ~0.8 bar of atmosphere (MAVEN)
- **Surface habitability:** limits the cosmic-ray flux and radiation dose at the surface, protects the ozone layer from charged particles; on Venus the missing field is one likely contributor to hydrogen escape
- **Geological record:** magnetic minerals record the field history (paleomagnetism)
- **Interior probe:** field properties constrain the deep interior
- Debated: how essential is a global field for a habitable surface?

Source: Lammer et al. (2003); Jakosky et al. (2018)



A global field as a shield: (a) a magnetised planet holds the solar wind off at the magnetopause, and the upper atmosphere lies inside the field lines; (b) at an unmagnetised planet the solar wind reaches the upper atmosphere directly, neutral atoms leave by sputtering and ions by pickup, the fate of Mars after its dynamo died about 4 Gyr ago. Course schematic.

The induction equation and the three requirements for a dynamo

$$\frac{\partial \vec{B}}{\partial t} = \underbrace{\nabla \times (\vec{v} \times \vec{B})}_{\text{advection: flow amplifies the field}} + \underbrace{\eta \nabla^2 \vec{B}}_{\text{diffusion: ohmic decay}}, \quad \eta = \frac{1}{\mu_0 \sigma}$$

1. **Electrically conducting fluid:** liquid iron alloy (terrestrial planets), metallic hydrogen (gas giants)
2. **Convection:** thermal buoyancy (core hotter than mantle) or compositional buoyancy (inner-core crystallisation releases light elements)
3. **Sufficient flow vigour:** advection must beat diffusion, measured by the magnetic Reynolds number $R_m \gtrsim 10\text{--}100$

(a) The induction equation

$$\frac{\partial \vec{B}}{\partial t} = \nabla \times (\vec{v} \times \vec{B}) + \eta \nabla^2 \vec{B}$$

advection

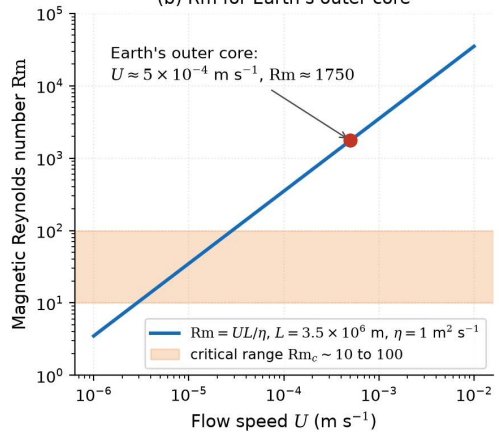
flow stretches and
folds field lines:
amplification, $\sim UB/L$

diffusion

ohmic resistance lets
the field decay,
 $\sim \eta B/L^2$

$$R_m = \frac{\text{advection}}{\text{diffusion}} = \frac{UL}{\eta}; \text{ a dynamo needs } R_m \gtrsim R_{m_c} \sim 10 \text{ to } 100$$

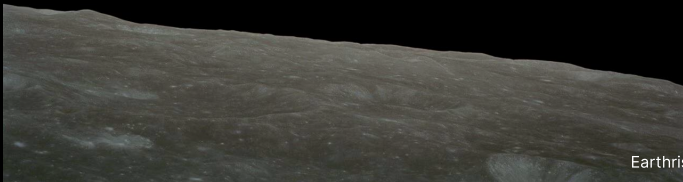
(b) R_m for Earth's outer core

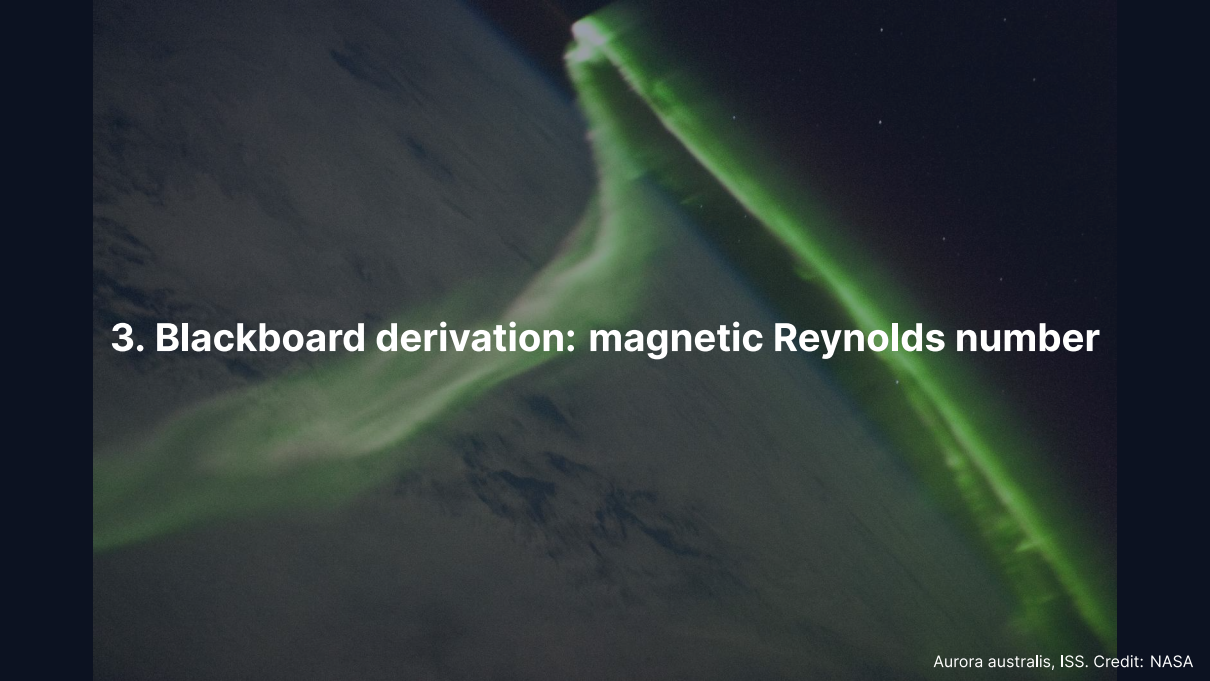


(a) The induction equation: advection amplifies the field at a rate $\sim UB/L$, diffusion lets it decay at $\sim \eta B/L^2$, and their ratio $R_m = UL/\eta$ must exceed the critical value, of order 10 to 100.

(b) R_m for Earth's outer core against flow speed ($L = 3.5 \times 10^6 \text{ m}$, $\eta = 1 \text{ m}^2 \text{ s}^{-1}$): at $U \approx 5 \times 10^{-4} \text{ m s}^{-1}$ the estimate is about 1750, far above the critical range. Course figure.

Break



A photograph of the Aurora australis (southern aurora) seen from the International Space Station (ISS). The aurora appears as a bright green, glowing arc against the dark background of the Earth's night sky. The arc is slightly curved and has a textured, shimmering appearance. The Earth's surface is visible in the lower part of the frame, showing dark, swirling patterns of clouds and landmasses.

3. Blackboard derivation: magnetic Reynolds number

Deriving the magnetic Reynolds number

Goal: from the induction equation, derive $R_m = UL/\eta$ and show that advection dominates in Earth's core. Dimensional analysis of each term: advection $|\nabla \times (\vec{v} \times \vec{B})| \sim UB/L$, diffusion $|\eta \nabla^2 \vec{B}| \sim \eta B/L^2$.

$$R_m = \frac{\text{advection}}{\text{diffusion}} = \frac{UB/L}{\eta B/L^2}, \quad \text{the field strength } B \text{ cancels:}$$

$$R_m = \frac{UL}{\eta}$$

- $R_m \gg 1$: advection dominates, field “frozen in” to the flow; $R_m \ll 1$: diffusion wins, field decays; critical $R_{m_c} \sim 10\text{--}100$
- R_m depends on the **flow**, not on the field

Application: Earth's outer core

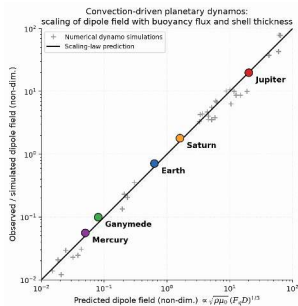
Parameter	Symbol	Value
Core radius	L	$3.5 \times 10^6 \text{ m}$
Flow velocity	U	$5 \times 10^{-4} \text{ m s}^{-1}$
Magnetic diffusivity	η	$\sim 1 \text{ m}^2 \text{ s}^{-1}$

$$Rm_{\oplus} = \frac{5 \times 10^{-4} \times 3.5 \times 10^6}{1} \approx 1750$$

$Rm \approx 1750 \gg Rm_c$: advection overwhelmingly dominates.

Ohmic diffusion timescale: $\tau_{\text{ohm}} \sim L^2 / \eta \approx 400,000 \text{ yr} \ll \text{Earth's age} \Rightarrow \text{field must be **continuously regenerated**}.}$

Dynamo scaling laws



Credit: Course figure, after Christensen & Aubert (2006)

Dynamo simulations collapse onto simple power laws:

- Field strength $\propto$ (convective power) $^{1/3}$
- **Independent of rotation rate**
- Predicts Earth, Jupiter, Mercury, Ganymede from first principles
- Works across ~4 orders of magnitude in parameters

$B_{\text{dip}} \propto \sqrt{\rho\mu_0} (F_q D)^{1/3}$ (F_q buoyancy flux, D shell thickness, ρ core density):
field strength is set by the **available convective power**, not by rotation rate; Venus is absent, its dynamo off.

Hf–W chronometry: beyond the first estimate

Modern refinements of the core formation timescale.

- Pressure-dependent partitioning: $D_W^{\text{met/sil}}$ decreases at high P
- At deep magma ocean pressures (~60 GPa), D_W is ~10 instead of ~40 (surface)
- This widens the permitted window to ~30–100 Myr, depending on how much metal re-equilibrates
- Complementary isotope systems (Pd–Ag, Pt–Os) cross-check the timing
- Evidence for **disequilibrium core formation**: not all metal re-equilibrates with silicate before sinking
- Earth's core likely formed in episodic accretion events, not continuous

Source: Rudge et al. (2010); Kleine et al. (2017)

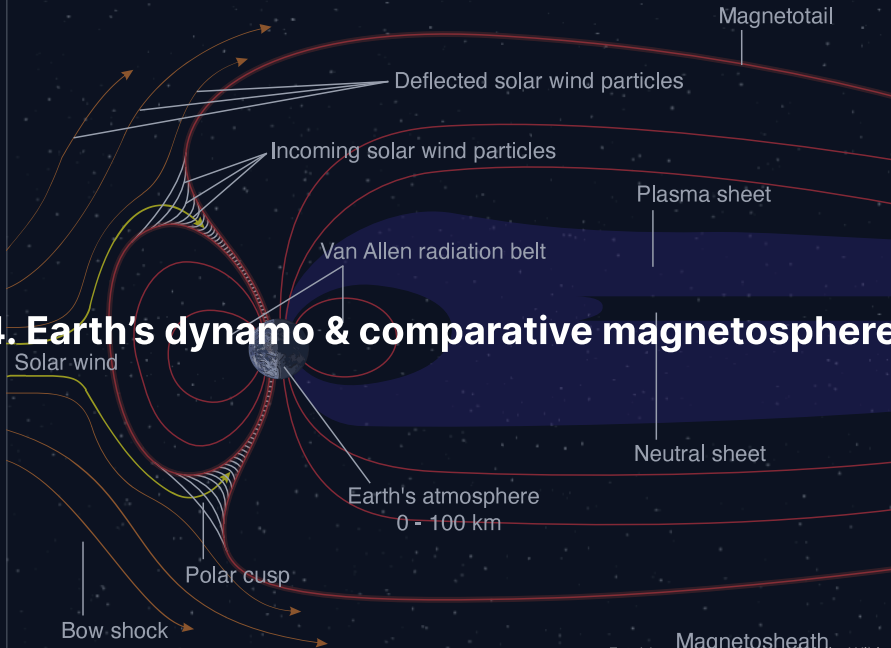
How old is Earth's magnetic field?

- Oldest confirmed field: ~3.4–3.5 Gyr (Archean South Africa, Tarduno 2010); single-crystal zircon paleomagnetism pushes this to 4.2 Gyr (Tarduno et al. 2015)
- Archean (>2.5 Ga): records in komatiites and banded iron formations; the field is continuous from ≥ 3.4 Ga to today
- Proterozoic (2.5–0.54 Ga): apparent polar wander paths trace plate motions and supercontinent cycles; Thellier paleointensities similar to or slightly above modern
- Some Proterozoic intervals show very frequent reversals followed by superchrons

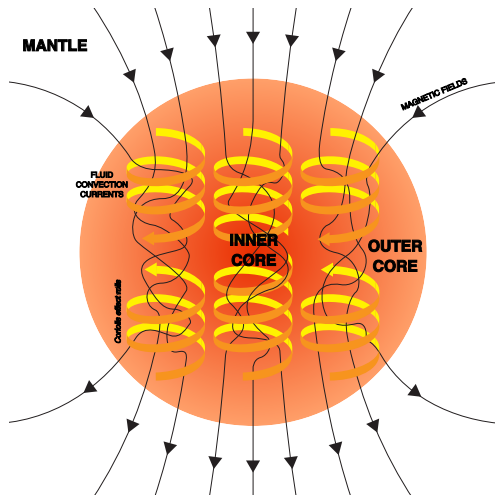
The dynamo was active during the Hadean eon, shielding Earth's atmosphere during the time when life was emerging.

Source: Tarduno et al. (2010, 2015); Biggin et al. (2015); Tarduno et al. (2020)

4. Earth's dynamo & comparative magnetospheres



Earth's geodynamo



Credit: Wikimedia, CC BY-SA 4.0

- **Outer core:** liquid Fe alloy (1220–3480 km from centre)
- Two driving mechanisms:
 1. Thermal convection (core hotter than mantle)
 2. Compositional convection (inner core crystallisation)
- Predominantly **dipolar** field
- Tilted $\sim 11^\circ$ from rotation axis
- Surface field: 25–65 μT

Earth's core: composition & growth

Inner core (0–1220 km):

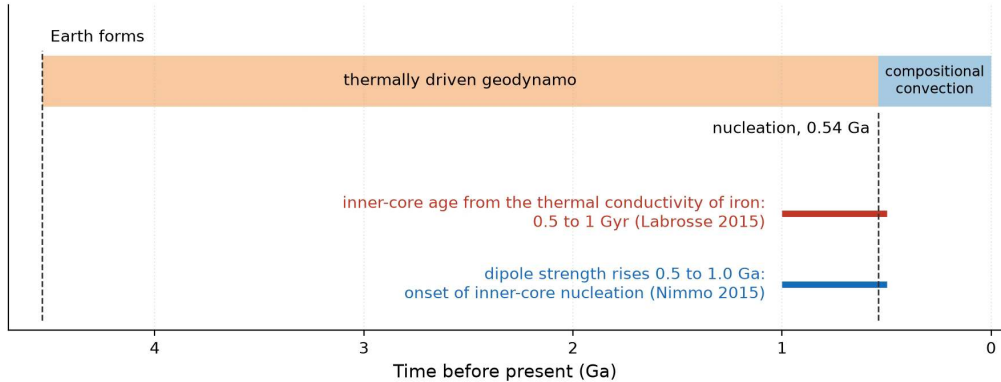
- Solid iron–nickel alloy
- Growing at ~0.5 mm/yr
- Age: ~0.5–1 Gyr (far younger than Earth)
- Latent heat of crystallisation powers the dynamo today

Outer core (1220–3480 km):

- Liquid Fe–Ni alloy + light elements
- ~5–10% less dense than pure Fe
- Light elements: S, Si, O, C, H
- Vigorous convection $\Rightarrow$ dynamo

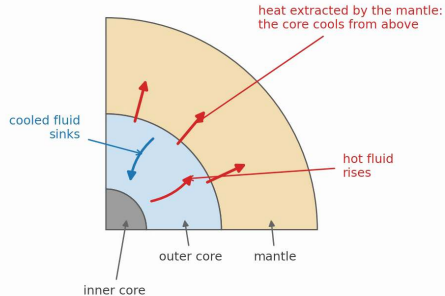
Compositional convection (inner core crystallisation ejects light elements) is the **dominant** driver of today's geodynamo.

When did Earth's inner core nucleate?

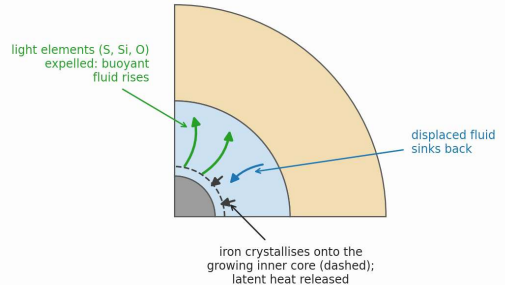


The inner-core age debate on a timeline: the thermal conductivity of iron gives an inner-core age of only 0.5 to 1 Gyr (Labrosse 2015), and the paleomagnetic dipole strength rises 0.5 to 1.0 Gyr ago, read as the onset of compositional convection at nucleation (Nimmo 2015); together they place nucleation near 0.54 Ga; before that the geodynamo ran on thermal convection alone. Course figure.

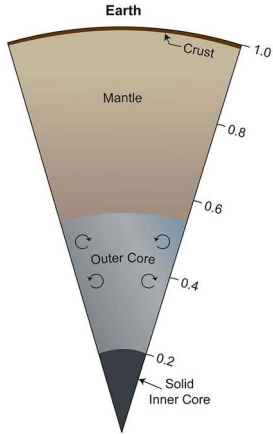
(a) thermal convection: the core cools



(b) compositional convection: the inner core grows

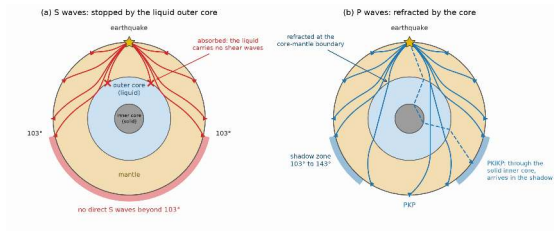


The two buoyancy sources of the geodynamo, in a quarter-section cutout (layer radii to scale): (a) the mantle extracts heat across the core-mantle boundary, so the core cools from above and convects; (b) inner-core growth rejects light elements (S, Si, O) and releases latent heat, driving compositional convection. Course figure.

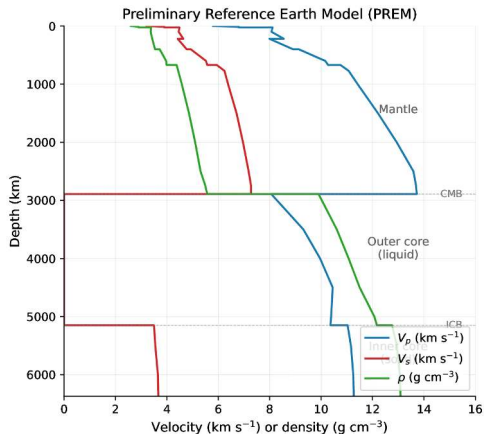


Cross-section of Earth's interior. Credit: NASA/JPL-Caltech/SwRI (PIA25063).

The shadow-zone argument



- S waves cannot cross a liquid: no direct S beyond 103° $\Rightarrow$ the outer core is **liquid** (Oldham 1906)
- V_p drops $\sim 13.7 \rightarrow 8$ km/s at the CMB: rays refract steeply $\Rightarrow$ P shadow zone from 103° to 143°
- Refracted core phases (PKP) emerge beyond 143°
- Phase letters spell the ray path, one per leg: P mantle, K liquid outer core, I solid inner core
- Weak arrivals *inside* the shadow (PKIKP): refracted again by a **solid inner core** (Lehmann 1936)



PREM seismic velocities and density against depth: V_s (red) vanishes between 2890 and 5150 km depth, so the outer core transmits no shear waves and is liquid; the density (green) jumps at the core-mantle boundary and again at the inner-core boundary, the metal-silicate and the solid-liquid interfaces. Seismology hands the dynamo its arena: a 2260 km thick shell of convecting liquid iron. Course figure, after Dziewonski & Anderson (1981).

Secular variation of the field

Earth's field is not static. It varies on many timescales:

- **Secular variation** (decades to centuries): non-dipole components grow, decay, drift
- **Westward drift**: non-dipole features migrate westward at $\sim 0.2^\circ/\text{yr}$, implying differential rotation between core and mantle
- **South Atlantic Anomaly (SAA)**: a $\sim 30\%$ reduction in field strength over South America and the South Atlantic
- SAA is the surface expression of a reversed-flux patch at the core–mantle boundary

Source: Finlay et al. (2020), Swarm satellite constellation data

Magnetic field reversals and superchrons

- N and S magnetic poles **swap**; current rate ~4–5 per Myr, each reversal ~1000–10 000 yr; the field weakens, becomes multipolar, re-establishes
- **Magnetic stripes** on the ocean floor: key evidence for seafloor spreading
- **Cretaceous Normal Superchron** (~121–83 Ma): one polarity for ~38 Myr; also a Permo-Carboniferous reversed superchron (~312–262 Ma)
- The reversal rate is modulated by core-mantle boundary heat flow, set by slow mantle convection, over hundreds of Myr

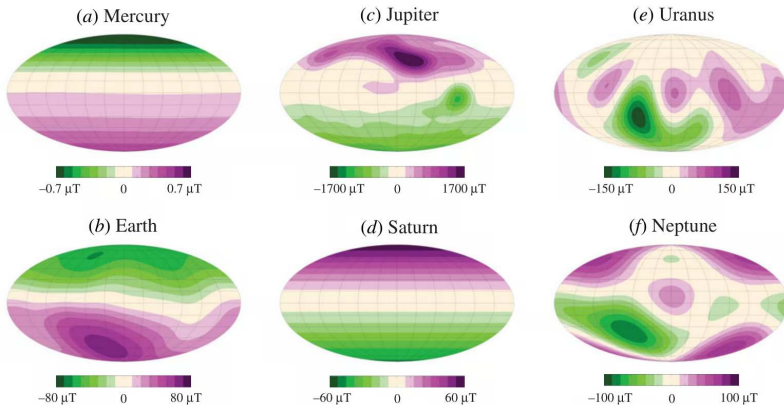
Credit: Course figure; chron ages after Gradstein et al. (2020)

Source: Biggin et al. (2012); Olson & Amit (2015)

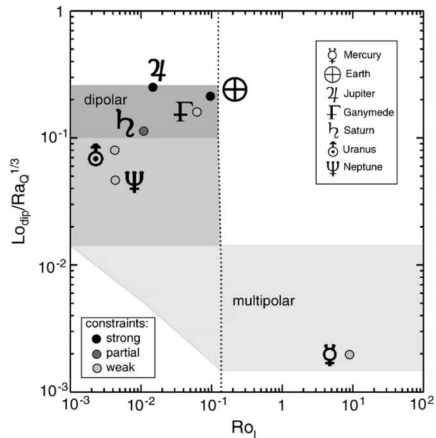
Magnetic fields across the solar system

Body	Field type	Surface field	Rel. dipole
Earth	Active dynamo	25–65 μT	1
Mercury	Active (weak)	$\sim 0.3 \mu\text{T}$	5×10^{-4}
Venus	None detected	$< 0.01 \mu\text{T}$	$< 10^{-5}$
Mars	Remnant crustal	up to 1500 nT	n/a
Jupiter	Active (strong)	400–1400 μT	20,000
Saturn	Active	$\sim 20 \mu\text{T}$	~ 600
Ganymede	Active	$\sim 0.7 \mu\text{T}$	1.5×10^{-3}

Jupiter's field is $\sim 20,000\times$ Earth's, generated in metallic hydrogen.



Surface radial magnetic fields of six solar-system dynamos. Earth, Jupiter and Saturn: dipoles roughly aligned with the rotation axis, Saturn almost perfectly axisymmetric. Uranus and Neptune: multipolar, tilted $\sim 59^\circ$ and $\sim 47^\circ$ and offset from the planet centre. Mercury: weak and offset ~ 500 km northward. The per-panel colour scales span four orders of magnitude. One mechanism, many faces: geometry and shell structure set what the dynamo shows at the surface. Soderlund & Stanley (2020).



Dipolar versus multipolar dynamo regimes: dimensionless dipole moment against the local Rossby number (inertia relative to rotation), dark region dipolar, light multipolar, calibrated on numerical dynamo runs. Earth, Jupiter, Saturn and Ganymede sit in the dipolar regime; Mercury, Uranus and Neptune lie in or near the multipolar one. Rapidly rotating, thick-shell dynamos stay dipolar. Olson & Christensen (2006).

Venus and Mercury: two puzzles

Venus: no field, though Earth's twin in size and composition

- Slow rotation (243-day period)? Simulations say this alone is insufficient
- No inner core: thermal convection alone may be too weak
- Stagnant lid: the mantle may not extract core heat fast enough

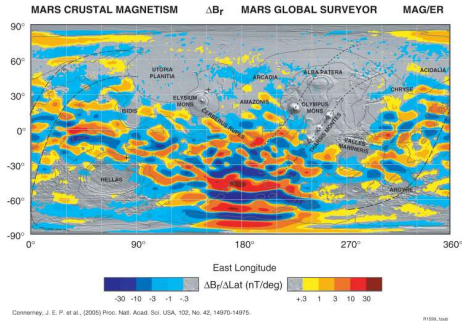
Mercury: an offset dynamo

- Surface field $\sim 0.3 \mu\text{T}$, 1% of Earth's; dipole-dominated, weak higher harmonics
- Offset ~ 500 km north of the geometric centre (MESSENGER, 2011–2015)
- A thin liquid shell above a partly solid deeper core limits the convective vigour

Source: Anderson et al. (2012)

Venus most likely combines reduced core cooling (no plate tectonics) with no inner-core nucleation.

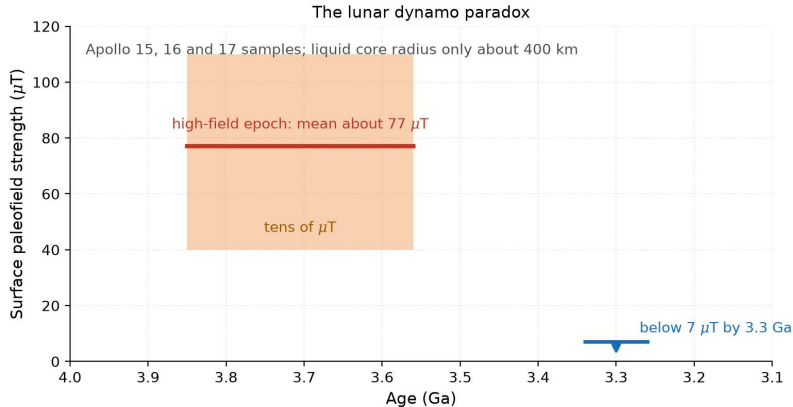
Mars: a fossil magnetic field



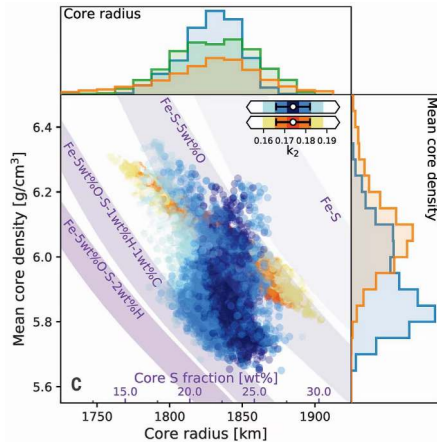
Credit: NASA/GSFC

- No global field today; intense **crustal magnetisation** in the southern highlands, absent in the young northern lowlands and the large basins
- Mars **had** an active dynamo; it died between ~4.1 and 3.7 Ga
- Magnetised ~3.7 Ga Lucus Planum lavas (Mittelholz et al. 2020): the dynamo was intermittent or shut down later than the classical ~4.1 Ga
- MAVEN: present-day loss of ~2–3 kg s⁻¹ (ions ~100 g s⁻¹); with the higher rates under the young, active Sun the 4 Gyr integral is ≥ 0.8 bar of CO₂

Source: Mittelholz et al. (2020); Jakosky et al. (2018)



The lunar dynamo paradox: surface paleofield strength from the Apollo 15, 16 and 17 samples against age. The high-field epoch from about 3.85 to 3.56 Ga has a mean of about 77 μT , and the field had dropped below 7 μT by about 3.3 Ga (Weiss & Tikoo 2014). A field of tens of microtesla from a liquid core of only about 400 km radius is difficult to generate by thermal convection alone. Course figure.



InSight's Mars core: radius and density posteriors from independent inversions, which converge on $R_{\text{core}} \approx 1830$ km and a mean core density near 6 g cm^{-3} . The purple bands (lines of fixed composition) demand several wt% of light elements: the core is too light for pure iron.

A sulfur-rich core may never nucleate an inner core: no compositional buoyancy, an early-dying dynamo. Stähler et al. (2021).

Jupiter and Ganymede: two more dynamos

Jupiter: the strongest field in the solar system, $\sim 20\,000\times$ Earth's dipole moment

- H_2 becomes **metallic** above ~ 100 GPa, at ~ 0.85 Jupiter radii; conductivity like a liquid metal
- Juno gravity: the transition is gradual, not sharp

Ganymede: the only moon with an intrinsic field

- Detected by Galileo (1996): surface field $\sim 0.7\ \mu\text{T}$, dipole tilted $\sim 176^\circ$
- Small ($R = 2634$ km), so expected thermally dead; tidal heating from the Laplace resonance or slow freezing of the outer core may sustain the convection
- JUICE (arriving 2031) will map the field in detail

Source: Connerney et al. (2022); Bolton et al. (2017); Kivelson et al. (1996); Bland et al. (2008)

Body	Dynamo today	What the field tells us
Mercury	 active	weak, about 1% of Earth's; dipole offset northward; thin liquid shell
Venus	 none detected	no inner core to drive compositional convection; stagnant lid slows core cooling
Earth	 active	reference case; inner-core nucleation within the past 1 Gyr
Moon	 extinct	paleofields of tens of microtesla from 3.85 to 3.56 Ga, below 7 microtesla by 3.3 Ga
Mars	 extinct	remnant crustal magnetism in the southern highlands; global field lost about 4 Gyr ago
Ganymede	 active	the only moon with a dynamo; surface field about 0.7 microtesla, above Mercury's 0.3

Dynamo status of the terrestrial bodies and Ganymede, the only moon with an active dynamo: Mercury weak and active, offset northward; Venus none detected, no inner core and a stagnant lid; Earth active, inner-core nucleation within the past 1 Gyr; the Moon extinct, with paleofields of tens of microtesla from 3.85 to 3.56 Ga; Mars extinct, remnant crustal magnetism, global field lost about 4 Gyr ago; Ganymede active, surface field about 0.7 μT against Mercury's 0.3 μT . Course figure.

Magnetopause standoff distance

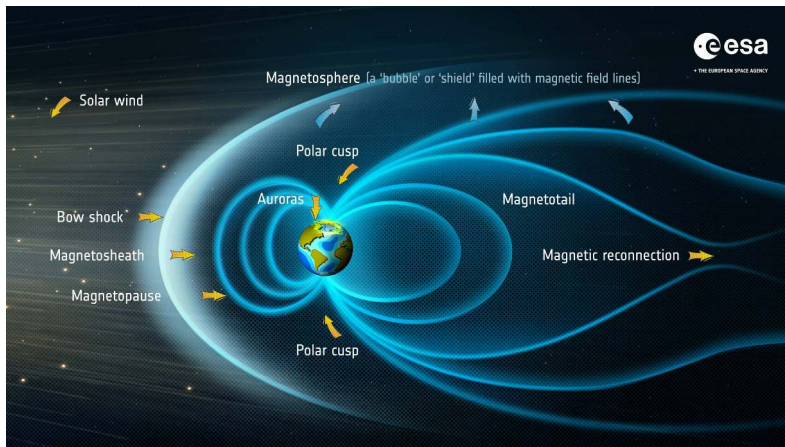
Pressure balance at the magnetopause:

$$\underbrace{\frac{1}{2} \rho_{\text{sw}} v_{\text{sw}}^2}_{\text{solar wind ram pressure}} = \underbrace{\frac{B_{\text{mp}}^2}{2\mu_0}}_{\text{magnetic pressure}}$$

For a dipole field, $B \propto r^{-3}$, so the standoff distance scales as:

$$r_{\text{mp}} \propto \left(\frac{B_0^2}{\mu_0 \rho_{\text{sw}} v_{\text{sw}}^2} \right)^{1/6}$$

- Earth under typical solar wind: $r_{\text{mp}} \approx 10 R_{\oplus}$
- During a CME: compressed to $\sim 6 R_{\oplus}$
- Jupiter (much stronger field): $\sim 50\text{--}100 R_J$



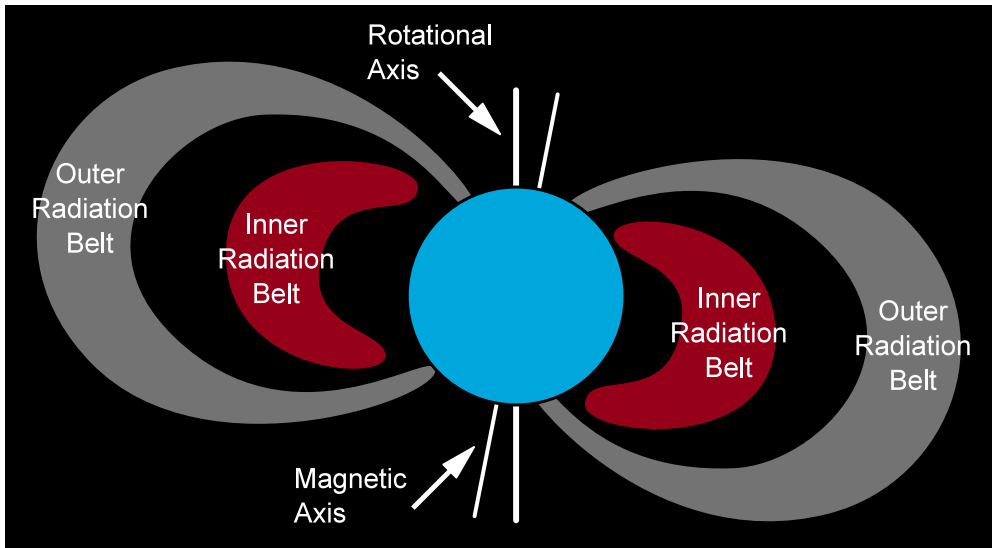
Anatomy of Earth's magnetosphere: the solar wind arrives from the left, the bow shock slows and deflects it, and the shocked plasma fills the magnetosheath. The magnetopause sits where ram pressure equals magnetic pressure, at $r_{mp} \approx 10 R_{\oplus}$ under typical solar wind; the compressed dayside field follows from that balance, and the solar wind flowing past stretches the nightside tail. Credit: ESA, CC BY-SA 3.0 IGO.

Magnetotail reconnection and the radiation belts

- On the night side the field is stretched into a tail ($>200 R_{\oplus}$); opposing field lines **reconnect** and release stored magnetic energy
- Earthward particles precipitate into the polar atmosphere (**aurorae**); downstream particles leave as **plasmoids** (self-contained blobs of magnetised plasma); reconnection drives **substorms** every few hours in active conditions
- **Van Allen belts** (Explorer 1, 1958): inner belt at $\sim 1.5 R_{\oplus}$ with 10–100 MeV protons from cosmic-ray interaction with the atmosphere (CRAND); outer belt at $4\text{--}5 R_{\oplus}$ with 0.1–10 MeV electrons injected during geomagnetic storms and accelerated by wave-particle interactions; a third, transient belt found by the Van Allen Probes (2012–2019); a major hazard for spacecraft and crews

Magnetic reconnection is the engine of space weather: it converts solar-wind kinetic energy into particle acceleration and auroral emission.

Source: de Pater & Lissauer (2010); Baker et al. (2013)



Cross-section of Earth's Van Allen radiation belts. Credit: Booyabazooka / NASA, public domain.

Aurorae: light from the magnetosphere



Credit: NASA/ISS

Solar wind particles channelled along field lines to polar regions:

- **Green:** excited O at 100–200 km (557.7 nm)
- **Red:** excited O at 200–400 km (630.0 nm)
- **Blue/violet:** excited N₂ bands

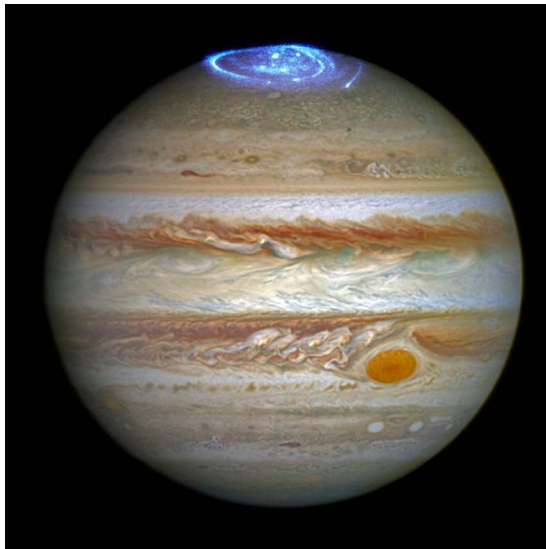
Observed on Jupiter, Saturn, and Uranus: any magnetised planet.

Why aurorae happen at the poles, and Jupiter's

- The dipole focuses incoming particles toward the magnetic poles; the polar **cusps** give the solar wind direct access to the atmosphere
- Electrons accelerated by tail reconnection precipitate down the field lines at 1–100 keV and reach 100–400 km altitude; the emission lines are fingerprints of the excited species
- **Jupiter:** 10^{13} – 10^{14} W of UV power, 10^3 – $10^4\times$ Earth's, powered by the ~ 10 h rotation and ~ 1 tonne s^{-1} of sulfur and oxygen from Io; Io, Europa and Ganymede leave **auroral footprints**

Aurorae map the power a magnetosphere dumps into its atmosphere: solar-wind coupling at Earth, rotation and Io's mass loading at Jupiter.

Source: Connerney et al. (2022); Mauk et al. (2017)



Jupiter's ultraviolet aurora over an optical image, Hubble Space Telescope. Credit: NASA, ESA, J. Nichols (University of Leicester).

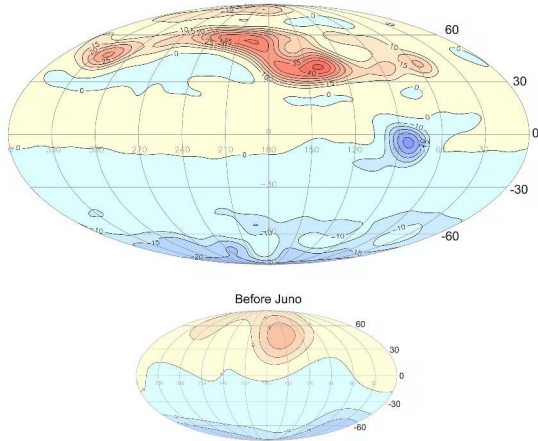
A photograph of the Aurora australis (Southern Lights) in a dark night sky. The aurora appears as a bright green, jagged, and elongated light structure. It has a main vertical column on the right side and a horizontal, diffuse band extending towards the left. The background is a deep black, with some faint, wispy green light visible on the left side.

5. Recent advances & outlook

Juno: Jupiter's asymmetric dynamo

- Juno (2016–) has measured Jupiter's gravity and magnetic field in unprecedented detail
- Surface field dramatically **non-dipolar**
- Northern hemisphere: intense concentrated “flux lobe”
- “Great Blue Spot”: a patch of reversed polarity near the equator
- Magnetic equator offset from the rotational equator
- Implies dynamo operates at relatively shallow depths in the stably-stratified dilute core (Lecture 8)

Source: Connerney et al. (2022), *Journal of Geophysical Research*



Jupiter's surface radial field after Juno's prime mission (top) and before Juno (bottom): yellow and red mark outward flux, blue inward, and the surface field is strongly non-dipolar. The Great Blue Spot is an intense localised patch of inward flux near the equator, invisible in the low-resolution pre-Juno map. The morphology places the dynamo in a deep but not razor-thin shell of metallic hydrogen. Credit: NASA/JPL-Caltech/SwRI (PIA25040).

Missions to cores and dynamos

- **BepiColombo** (ESA/JAXA, launched 2018): flybys 2021–2025 confirmed the weak, active, offset dipole; the two orbiters (MPO, MMO) will map it from polar orbit after the late-2026 orbit insertion
- **JUICE** (launched 2023, Jupiter 2031, Ganymede orbit 2034): separates Ganymede's intrinsic field from the induced ocean signature
- **MAVEN** (2014–): Mars loses $\sim 2\text{--}3 \text{ kg s}^{-1}$, mostly H and photochemically escaping O, faster during solar storms; over 4 Gyr about 0.8 bar of CO_2 , the link from a dead dynamo to a thin atmosphere (Lecture 5)
- **Psyche** (launched 2023, arrival 2029): a $\sim 220 \text{ km}$ M-type asteroid of bulk density $\sim 4 \text{ g cm}^{-3}$, a candidate exposed planetesimal core; its magnetometer tests for remnant magnetisation

Source: Benkhoff et al. (2021); Grasset et al. (2013); Jakosky et al. (2018); Elkins-Tanton et al. (2017)

Open questions in planetary magnetism

- **When did the inner core nucleate**, and how did Earth sustain a dynamo before it? (most estimates 0.5–1 Gyr ago)
- **Why does Venus have no detectable field?** Inner core, rotation, or tectonics?
- **Why is Mercury's field so weak yet active**, and what drives Ganymede's?
- **How do Uranus and Neptune generate multipolar fields?**
- **Will Earth reverse soon?** No reliable prediction method exists
- **How important are magnetic fields for habitability?**

Looking ahead to Lecture 5

This lecture ended with a planet that has a core, a mantle, a magnetic field, and a freshly outgassed secondary atmosphere.

- Lecture 5 (dr. Mara Attia) picks up that atmosphere: primary, secondary, and tertiary atmospheres
- Pressure and temperature with height: hydrostatic equilibrium
- The greenhouse effect and planetary energy balance
- How atmospheres are lost to space; magnetic shielding is one control on escape, but thermal escape proceeds regardless

Differentiation builds the planet; the atmosphere it outgasses is where climate and habitability are decided.

Summary

1. Magma oceans drive **metal–silicate separation**: iron sinks, silicates float
2. Hf–W chronometry gives a two-stage age of ~30 Myr for Earth's core formation, a lower limit; protracted accretion and partial equilibration place completion within ~100 Myr
3. Magma ocean crystallisation creates chemically distinct layers (crust, mantle reservoirs)
4. Dynamos require: conducting fluid + convection + sufficient R_m (≥ 10 –100)
5. Earth's $R_m \approx 1750$: vigorous dynamo action
6. Mars lost its dynamo between ~4.1 and 3.7 Ga (possibly intermittent, Mittelholz 2020) → atmospheric erosion
7. Venus has no detectable field: likely a combination of stagnant-lid tectonics, slow rotation, and/or absence of an inner core

Questions?

Next: Lecture 5, Atmospheres I: composition, structure & energy balance (Mara Attia)

Mon 21 Sep, 15:00–17:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 25 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

<https://ips.formingworlds.space>